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ORIGINAL ARTICLE

Treatment effects of high-dose antithrombin without concomitant heparin in patients with severe sepsis with or without disseminated intravascular coagulation

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Summary. *Background:* Disseminated intravascular coagulation (DIC) is a serious complication of sepsis that is associated with a high mortality. *Objectives:* Using the adapted International Society on Thrombosis and Haemostasis (ISTH) diagnostic scoring algorithm for DIC, we evaluated the treatment effects of high-dose antithrombin (AT) in patients with severe sepsis with or without DIC. *Patients and Methods:* From the phase III clinical trial in severe sepsis (KyberSept), 563 patients were identified (placebo, 277; AT, 286) who did not receive concomitant heparin and had sufficient data for DIC determination. *Results:* At baseline, 40.7% of patients (229 of 563) had DIC. DIC in the placebo-treated patients was associated with an excess risk of mortality (28-day mortality: 40.0% vs. 22.2%, $P < 0.01$). AT-treated patients with DIC had an absolute reduction in 28-day mortality of 14.6% compared with placebo ($P = 0.02$) whereas in patients without DIC no effect on 28-day mortality was seen (0.1% reduction in mortality; $P = 1.0$). Bleeding complications in AT-treated patients with and without DIC were higher compared with placebo (major bleeding rates: 7.0% vs. 5.2% for patients with DIC, $P = 0.6$; 9.8% vs. 3.1% for patients without DIC, $P = 0.02$). *Conclusions:* High-dose AT without concomitant heparin in septic patients with DIC may result in a significant mortality reduction. The adapted ISTH DIC score may identify patients with severe sepsis who potentially benefit from high-dose AT treatment.

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Introduction

Patients with severe sepsis usually present with generalized activation of the innate immune response that is capable of triggering coagulation and inhibiting fibrinolysis [1–4]. Hemostatic abnormalities associated with sepsis may vary from subtle laboratory evidence of coagulation activation to clinically overt, severe disseminated intravascular coagulation (DIC) [3]. The pathogenesis of sepsis-related DIC involves inflammation-induced excessive thrombin generation with loss of localization, degradation and dysfunction of anticoagulant pathways, impaired fibrinolysis, derangement of microvascular endothelium and consequently intravascular fibrin deposition [1,3,5]. Clinically, severe DIC may manifest as thromboembolic disease and contribute to multiple organ dysfunction by microvascular thrombotic complications [1,3,4]. Consumption of platelets and procoagulant factors secondary to massive and ongoing activation of the hemostatic system can also lead to a severe hemorrhagic disorder. This clinical manifestation of DIC is less frequent in sepsis, but may dominate the presentation of DIC in the setting of trauma or surgery [1,3–6].

A number of smaller clinical studies have suggested that sepsis-related DIC is associated with a high mortality, while attenuation of DIC may ameliorate organ dysfunction and possibly reduce mortality [7–12]. However, past studies on the prognostic impact of DIC in sepsis as well as interventional trials have been hampered by a lack of consensus criteria for the diagnosis and severity scoring of DIC. In an effort to standardize the approach to DIC, the Scientific Subcommittee on DIC of the International Society on Thrombosis and Haemostasis (ISTH) has proposed scoring templates for overt and non-overt DIC [5]. These templates have been validated prospectively in independent patient populations, both as a tool to define diagnostic scores for DIC and as a method to demonstrate prognostic associations for DIC with mortality risk [13,14]. They offer new perspectives for objective assessment of therapeutic interventions in DIC [15].

Antithrombin (AT) is the most abundant anticoagulant circulating in plasma [16]. The decrease of its plasma activity at onset of severe sepsis correlates with DIC and lethal outcome [7,17–19]. Furthermore, earlier clinical studies had suggested that AT administration might be beneficial in patients with DIC from severe sepsis or septic shock [9–12]. However, a large, multinational, randomized placebo-controlled trial in 2314 patients with sepsis (the KyberSept trial) found no benefit, in terms of mortality, from AT administration and there was an increased risk of bleeding in those who received both AT and heparin [20]. Of note, there was a trend towards a survival benefit in patients who received AT without concomitant heparin [16,20]. In line with this clinical evidence, studies on AT-heparin interactions *in vitro* and in animal models have meanwhile demonstrated that soluble glycosaminoglycans and heparin antagonize the anti-inflammatory and microcirculatory effects of AT in the setting of sepsis [16,21].

The following analysis based on the KyberSept database explores the treatment effect of AT in patients with severe sepsis who had DIC according to the ISTH scores. For the reasons stated above, this analysis is limited to patients not concomitantly treated with heparin [20]. We report that the use of AT without concomitant heparin in patients suffering from severe sepsis and DIC may result in a clinically meaningful and highly significant reduction in mortality. The adapted ISTH DIC score may be very meaningful in identifying a population of septic patients for further study who might potentially benefit from high-dose AT treatment.

Methods

The KyberSept study was a double-blind, prospectively randomized, placebo-controlled, multi-center phase 3 clinical trial in patients with severe sepsis. The study enrolled a total of 2314 patients in the intention-to-treat (ITT)-population who were randomized into two equal groups of 1157 to receive either intravenous high-dose AT (30 000 IU in total over 4 days) or placebo (1% human albumin, see reference 20 for more details). The patient population of the present analysis was restricted to all patients who did not receive concomitant heparin during the 4 days of study treatment and for whom baseline routine and central laboratory data for determination of DIC were available.

Laboratory methods

Blood samples were drawn pre-infusion and on study day 1. Citrated plasma and serum samples were stored at −70 °C until analyzed by the central laboratory. Platelet counts, prothrombin time (PT)/international normalized ratio (INR)/Quick, activated partial thromboplastin time (APTT), and AT levels were obtained at each time point. The central laboratory determined D-dimer from citrate-anticoagulated blood samples obtained prior to study drug infusion.

Definition of DIC

For definitions and scoring of non-overt and overt DIC see appendices 1 and 2. DIC criteria and scores were adopted as proposed by Taylor *et al.* [5] for platelets, PT and fibrinogen level. D-dimer values were chosen as fibrin-related marker and scored according to the cut off levels and ranges recently published by Levi *et al.* [15]. In addition to PT values, Quick test and PT ratio/INR were introduced into the adapted score with cut off points as indicated in appendices 1 and 2. PT, Quick or PT ratio/INR were used depending on availability from the KyberSept study records.

Criteria for non-overt DIC also included changes of platelet counts and PT (PT, Quick or PT ratio/INR) values over time [5]. For the assessment of changes over time, baseline and 24-h values were compared and changes were scored as published [5]. If 24-h values were not available, values recorded up to day 4 were accepted for comparison. Normal ranges for baseline AT and protein C were ≥ 70%.

Patients were classified as having DIC if they fulfilled the criteria for either non-overt DIC, overt DIC or both.

Definition of major bleeding

In the KyberSept trial, bleeding was classified as major if it was intracranial or required a transfusion of at least three units of packed red blood cells within 24 h.

Statistical analyses

All calculations were performed with SAS version 8.0. The following analyses had been defined prior to unblinding of the original study: 28-, 56- and 90-day landmark mortality analyses in the overall ITT population and in subgroups of concomitant heparin use; a Mantel-Haenszel test, stratified for the severity of the disease, or Fisher's exact test (unstratified) for the ITT population or the subgroups were performed and 95% CI for the treatment effect (risk ratio AT/placebo) calculated (see ref. 20 for details). Subgroup findings for mortality underwent multi-factorial logistic regression and were tested for significant interaction with the treatment group.

Additional statistical analyses for the present report were exploratory in nature. For these *post hoc* analyses, the indicated nominal *P*-values cannot serve to confirm the difference between treatment groups. They are provided as a descriptive tool to measure the level of evidence for each difference and to contribute to the generation of new working hypotheses. The resulting P-values were not corrected for the multiplicity of tests as these analyses are not intended to be utilized as confirmatory evidence for primary hypothesis testing.

Baseline comparisons between treatment groups were performed with two-sided t-tests for continuous data, Fisher's exact tests for binomial data or chi-square tests for categorical data (ITT). Landmark mortality data (28- and 90-day), the percentage of patients with bleeding events, and landmark mortality for patients with bleeding were compared with Fisher's exact tests (two-sided), risk ratios and their 95% CI between treatment groups [22]. Kaplan–Meier estimates were used to illustrate survival trends in patient groups receiving AT or placebo. The curves were compared with two-sample Wilcoxon tests generalized for censored data.

Results

Study population and DIC at baseline

From the subgroup of patients receiving study medication without concomitant heparin (698 patients: placebo, 346; AT, 352), central laboratory data were available for 599 patients (placebo, 294; AT, 305). For 563 of these patients (placebo, 277; AT, 286), sufficient PT and platelet count data were available both at baseline and at 24 h for the determination of DIC.

At baseline, 223 patients had non-overt and 42 overt DIC, with 36 patients fulfilling the criteria for both non-overt and overt DIC according to the adapted ISTH scores (see Appendices). Thus, a total of 229 of 563 sepsis patients (40.7%) represented the DIC population for the present analysis.

Baseline characteristics by treatment assigned and DIC status

Baseline data for the current study patients randomized to either the AT or the placebo group of KyberSept are given in Table 1. No significant differences were observed between the treatment groups with respect to age, gender, body weight, underlying diseases, site of infection, surgical status, percentage of patients with baseline AT levels below 60%, circulatory shock, and simplified acute physiology score (SAPS II) values. Significant differences between the AT and the placebo group were seen for ethnicity and blood culture results, but these differences did not affect mortality risk as predicted by SAPS II.

As expected, the percentage of patients with decreased baseline AT and the probability of dying according to the SAPS II score [23] tended to be lower in the group without DIC.

Prognostic impact of DIC

Placebo-treated patients with DIC at baseline were at higher risk of dying than patients without laboratory evidence of DIC. This was true for mortality rates at 28 days (40.0% vs. 22.2%; $P = 0.004$) and at 90 days (50.4% vs. 32.9%; $P = 0.002$; see Table 2). Over both treatment groups, there were weak positive correlations between SAPS II scores and the number of criteria fulfilled for the diagnosis of non-overt ($r = 0.23$; $P < 0.0001$) or overt DIC ($r = 0.09$; $P = 0.055$). In this subgroup of patients, we were unable to demonstrate an independent prognostic value of DIC in relation to sepsis mortality when investigated by a multifactorial logistic regression model with SAPS II as the factor with the highest prognostic value.

Efficacy of AT treatment in sepsis patients with DIC

The treatment effect of high-dose AT without concomitant heparin on all-cause mortality of patients presenting with severe sepsis and DIC is shown in Fig. 1 and Table 2. In DIC patients, Kaplan–Meier survival functions until day 90 were significantly different between AT and placebo groups ($P = 0.007$; Fig. 1A), whereas no effect of high-dose AT on survival was seen in patients without DIC ($P > 0.2$; Fig. 1B).

Among patients without DIC, 38 of 172 (22.1%) had died by day 28 in the AT group and 36 of 162 (22.2%) in the placebo group [risk ratio (95% confidence interval), 0.99 (0.66–1.49), $P > 0.2$]; among patients with DIC, 28-day mortality rates were 25.4% (29 of 114) with AT and 40.0% (46 of 115) with placebo [0.64 (0.43–0.94), $P = 0.024$]. By 90 days the mortality rates were 50 of 167 (29.9%) with AT and 51 of 155 (32.9%) with placebo [0.91 (0.66–1.26), $P > 0.2$] for patients without DIC, and 37 of 108 (34.3%) with AT and 58 of 115 (50.4%) with placebo [0.68 (0.49–0.93), $P = 0.015$] in patients with DIC, respectively. Absolute mortality reduction by high-dose AT treatment of severe sepsis in patients with DIC was 14.6% and 16.2% at 28 and 90 days, respectively.

Patients with overt DIC ($n = 42$) who received high-dose AT without concomitant heparin were found to have a trend towards a reduced mortality rate at 28 days compared with placebo patients: four of 20 (20.0%) vs. eight of 22 (36.4%), respectively [0.55 (0.20–1.55), $P > 0.2$]; at 90 days, mortality rates were six of 20 (30.0%) with AT and nine of 22 (40.9%) with placebo [0.73 (0.32–1.69), $P > 0.2$].

For patients who met the criteria for non-overt DIC ($n = 223$), both, the 28- and the 90-day mortality rates were significantly reduced in the AT group with a > 30% relative risk reduction compared with placebo. At 28 days, 29 of 113 (25.7%) of the AT patients not receiving concomitant heparin had died in contrast to 45 of 110 (40.9%) of the placebo patients [0.63 (0.43–0.92), $P = 0.023$]. At 90 days, the absolute treatment effect was even stronger with a mortality rate of 37 of 107 (34.6%) for the AT group and 57 of 110 (51.8%) for the placebo group [0.67 (0.49–0.92), $P = 0.014$].

Safety of AT and no heparin in DIC

Major bleeding rates were analyzed for the study drug infusion period (4 days) plus 24 postinfusion days. In patients without DIC at baseline, AT treatment was associated with a significant

Table 1 Baseline parameters of patients with DIC (either non-overt or overt) or without DIC receiving either placebo or high-dose AT without concomitant heparin

	Non-overt DIC			Overt DIC			No DIC		
	Placebo (n = 110)	AT (n = 113)	P-value	Placebo (n = 22)	AT (n = 20)	P-value	Placebo (n = 162)	AT (n = 172)	P-value
Age (years), mean (± SD)	56.1 (17.2)	55.8 (17.7)	0.90*	53.6 (16.1)	55.9 (18.5)	0.67*	54.9 (17.2)	55.5 (17.3)	0.73*
Gender (% male)	68.2	61.1	0.33†	59.1	65.0	0.76†	57.4	62.2	0.37†
Body weight (kg), mean (± SD)	76.6 (19.4)	73.2 (17.3)	0.19*	81.7 (14.4)	72.3 (14.4)	0.05*	75.8 (20.9)	77.1 (20.5)	0.60*
Ethnic group (%)			0.51‡			0.64‡			0.03‡
White	74.6	80.5		81.8	70.0		66.7	79.1	
Black	14.6	12.4		13.6	20.0		19.8	10.5	
Other	10.9	7.1		4.6	10.0		13.6	10.5	
Underlying disease or site of infection§ (%)			1.00‡			0.15‡			0.76‡
Respiratory	29.1	27.4		31.8	20.0		33.3	37.8	
Intra-abdominal infection	20.9	21.2		13.6	5.0		13.0	15.1	
Genitourinary	10.0	9.7		4.6	0.0		11.7	8.7	
Injury and poisoning	8.2	8.9		0.0	20.0		9.3	7.6	
Other	31.8	32.7		50.0	55.0		32.7	30.8	
Blood culture results (%)			0.02*			0.93*			0.03*
Gram-negative	10.9	22.1		22.7	25.0		15.4	12.8	
Gram-positive	21.8	17.7		9.1	15.0		19.1	11.1	
Other/mixed	1.8	7.1		4.6	5.0		0.6	4.1	
Not done/not verified	65.5	53.1		63.6	55.0		64.8	72.1	
Surgical patients (%)	40.9	41.6	1.00†	36.4	25.0	0.51†	40.1	37.2	0.65†
Baseline antithrombin < 60% (%)	69.1	71.7	0.77†	81.8	75.0	0.71†	31.3	26.3	0.37†
Circulatory shock (%)	53.6	51.3	0.55†	45.5	70.0	0.13†	40.1	47.7	0.38†
Mortality risk probability by SAPS II, customized for sepsis [23], mean (± SD)	0.51 (0.21)	0.47 (0.22)	0.29*	0.47 (0.21)	0.56 (0.21)	0.15*	0.39 (0.22)	0.40 (0.22)	0.77*

Treatment groups were compared by the following tests: *2-sided t-test, †two-sided Fisher's exact test, ‡chi-square test for independence of treatment group and levels of factor, §respiratory system, genitourinary system, injury and poisoning according to main chapters of International Classification of Diseases, 9th revision. Intra-abdominal infection includes the following ICD codes of main chapter 'Digestive system': acute appendicitis with generalized peritonitis and abscess, all ulcers with perforation excluding ulcer of esophagus, abscess of liver, acute cholecystitis and pancreatitis, cholangitis, other cholecystitis, abscess of intestine, diverticulitis of colon, diverticulitis of small intestine, other specified peritonitis, other suppurative peritonitis, perforation of intestine, peritonitis, peritonitis in infectious diseases, unspecified peritonitis.

Table 2 Twenty-eight and 90-day mortality for patients with or without DIC receiving high-dose AT or placebo without concomitant heparin

Subgroup	Days	Total number of patients	AT: % died	Placebo: % died	RR AT/ placebo	95% CI lower limit	95% CI upper limit	P-value (Fisher)
DIC	28	229	25.4	40.0	0.64	0.43	0.94	0.024
No DIC	28	334	22.1	22.2	0.99	0.66	1.49	> 0.2
DIC	90	223	34.3	50.4	0.68	0.49	0.93	0.015
No DIC	90	322	29.9	32.9	0.91	0.66	1.26	> 0.2
Non-overt DIC	28	223	25.7	40.9	0.63	0.43	0.92	0.023
No non-overt DIC	28	340	22.0	22.2	0.99	0.66	1.48	> 0.2
Non-overt DIC	90	217	34.6	51.8	0.67	0.49	0.92	0.014
No non-overt DIC	90	328	29.8	32.5	0.92	0.66	1.26	> 0.2
Overt DIC	28	42	20.0	36.4	0.55	0.20	1.55	> 0.2
No overt DIC	28	521	23.7	29.0	0.82	0.61	1.09	0.196
Overt DIC	90	42	30.0	40.9	0.73	0.32	1.69	> 0.2
No overt DIC	90	503	31.8	40.3	0.79	0.62	1.00	0.051

increase in the risk of major bleeding (9.8%) compared with a rate of 3.1% ($P < 0.05$) in placebo-treated patients (Table 3). Remarkably, in patients with DIC at baseline, the major bleeding rate was 7.0% in AT patients and did not significantly differ from the major bleeding risk (5.2%) found in the placebo patients (Table 3). The odds of major bleeding were not significantly different between patients with or without DIC in the AT treatment group (Breslow–Day interaction test, $P = 0.23$).

Discussion

Earlier clinical studies indicate that DIC may be an important predictor of mortality and that amelioration of DIC is

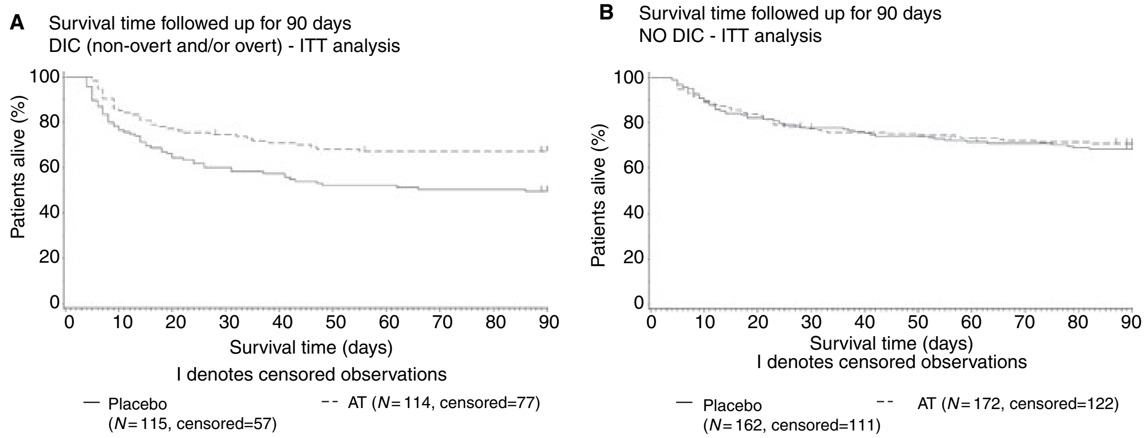

Fig. 1. Kaplan–Meier plots of patients with (A) or without (B) DIC receiving antithrombin (AT) or placebo and no concomitant heparin.

Table 3 Bleeding incidence of patients with and without DIC in the AT and placebo groups without concomitant heparin

Subgroup	Severity	Placebo	AT	Risk ratio AT/placebo, (95% CI)	Nominal *P*-value (Fisher's exact test, two-sided)
DIC	Number of patients	115	114		
	Minor*	6.1%	14.9%	2.45 (1.06–5.68)	0.03
	Major†	5.2%	7.0%	1.35 (0.48–3.75)	0.6
	Any bleeding	11.3%	18.4%	1.63 (0.86–3.10)	0.14
No DIC	Number of patients	161‡	173‡		
	Minor*	6.2%	14.0%§	2.25 (1.11–4.55)	0.03
	Major†	3.1%	9.8%	3.16 (1.20–8.38)	0.02
	Any bleeding	8.7%	20.2%	2.33 (1.30–4.16)	0.003
Total	Number of patients	276‡	287‡		
	Minor*	6.2%	14.3%¶	2.33 (1.36–4.00)	0.001
	Major†	4.0%	8.7%	2.19 (1.10–4.36)	0.025
	Any bleeding	9.8%	19.5%	2.00 (1.30–3.06)	0.001

*Within 14 days after start of treatment; †within 28 days after start of treatment; ‡as treated analysis; §refers to 172 non-missing values; ¶refers to 286 non-missing values. Breslow–Day test for homogeneity of treatment – differences between subgroups: $P > 0.2$ for minor, major and any bleeding.

associated with an improvement of organ function [7–12]. The present analysis suggests that the use of AT without concomitant heparin in patients suffering from severe sepsis and acute DIC results in a significant mortality reduction. This treatment effect appears to be independent of the severity of DIC as a survival benefit was also evident in the subgroup of patients with non-overt DIC. Given the limitations of *post hoc* subgroup analyses, the strength of this observation lies in its biological plausibility and that it is congruous with the findings in a series of previous studies with the use of AT in DIC [9–12]. The clinical relevance of these studies were questioned after the large KyberSept trial turned out to be negative in a patient population that included patients with severe sepsis both with or without DIC, and with or without heparin therapy [20]. Although patients in this trial were not stratified for DIC and use of heparin, baseline characteristics in the present analysis were evenly distributed between the treatment and control groups. Data suggest that beneficial effects of AT in the treatment of DIC may well be seen when the intrinsic patient heterogeneity that is typically observed in large sepsis populations, such as in the KyberSept trial, is reduced.

One of the putative options in the treatment of DIC, in addition to supportive critical care and blood products, has been the administration of heparin, but clinical reports have not uniquely demonstrated efficacy of heparin use in septic patients with DIC [24–26]. Avoidance of heparin may be preferable or even beneficial when AT is given for the treatment of severe sepsis [20]. As a result of known undesirable drug–drug interactions between heparin and AT [16,21], the present analysis focused on patients from the KyberSept trial who were treated with AT without receiving concomitant heparin. In this group of patients with severe sepsis, AT showed significant beneficial effects in the presence of DIC while it was inactive in the absence of DIC. Thus, results of this *post hoc* analysis from this large multicenter trial support conclusions drawn from previous investigations on much smaller study populations with the use of AT in DIC.

In 1978, Schipper *et al.* [27] first reported the use of AT transfusion in DIC. In 1985, Blauhut *et al.* [9] presented the results of a randomized study of AT vs. heparin in 51 shock patients with DIC. Duration of DIC was shortest for patients receiving AT, followed by patients who were treated with AT and heparin and patients who received heparin only. Clinical studies soon followed that investigated the potential beneficial effect of AT in patients with sepsis, with or without DIC [10–12,28]. In one randomized, placebo-controlled, double-blind study in patients with septic shock and DIC, high-dose AT resulted in a 44% reduction in mortality and duration of DIC was significantly reduced in the AT group [10]. In that study, concomitant heparin administration was not given during the 4-day AT supplementation period or over the entire 28-day study period except for low molecular weight heparin (enoxaparin 0.5 mg kg^{-1} given subcutaneously) for vascular catheter patency for those patients receiving extra renal replacement therapy.

Disseminated intravascular coagulation is a common complication of inflammatory diseases [1–4] and is associated with the systemic inflammatory response syndrome (SIRS) in up to 83% of patients [8]. In a retrospective study on 118 patients, generalized infection was the most frequent identifiable cause of DIC [29]. However, objective diagnosis of DIC in earlier studies was limited by the lack of standard criteria such as those recently proposed by the ISTH subcommittee on DIC [5]. A prospective validation of the ISTH scoring system for overt DIC revealed an incidence of 32% in intensive care patients associated with a 28-day mortality of 45% [13]. In another prospective assessment of the ISTH scoring templates in intensive care patients, non-overt DIC occurred in 20% of the patients with a 28-day mortality of 77.8%. Overt DIC was diagnosed in only 11% of these patients and was associated with a remarkably similar mortality of 77.6% [14]. In a retrospective analysis of the database from the PROWESS clinical trial on the use of drotrecogin alfa (activated) in severe sepsis, the incidence of overt DIC at study entry was 29% with a 28-day mortality of placebo-treated DIC patients of 43% [15]. Hence, despite standardization of criteria, the incidence and severity of DIC varies considerably among different patient cohorts studied and it remains undetermined whether distinguishing between non-overt and overt DIC according to the ISTH scores clearly separates subgroups of DIC patients with different prognosis.

In the present investigation of KyberSept patients with severe sepsis not receiving heparin during study drug infusions and for whom central laboratory data were available, a total of 41% (229 of 563) of patients had DIC at baseline associated with a 28-day mortality rate of 40% in the placebo group. In line with the findings referenced above, 28-day mortality rates in the placebo group were quite similar in patients with non-overt and overt DIC (40.9% and 36.4%, respectively). While this observation of similar outcomes in sepsis-induced overt vs. non-overt DIC is consistent with other clinical studies [14], the number of patients with overt DIC in the present analysis was quite small ($n = 42$) making it difficult to draw firm conclusions on the prognostic implications on clinical endpoints or treatment differences in sepsis patients based upon these laboratory definitions of DIC. However, it is important to note that a clear beneficial effect of AT was seen in the comparatively large group of sepsis patients ($n = 223$) with non-overt DIC. These results suggest that it may be appropriate to identify and treat DIC in patients with severe sepsis already at a non-overt or 'compensated' stage of coagulopathy.

On the contrary, it has not been satisfactorily demonstrated that non-overt DIC is truly indicative of a 'compensated' state. It may very well depend on the natural history of the inflammatory and coagulopathic state in each individual patient at the time the patient is initially assessed. Some patients may be first evaluated at an early point in a progression of microangiopathy, which is destined to accelerate over time to overt DIC. In others, the diagnosis of non-overt DIC may indeed remain in a truly compensated state. Cognizance of this still-unknown and uncharacterized difference may warrant careful thought in interpretation of future therapeutic trials. Differentiation may require more subtle markers of inflammation/hemostasis upregulation than have been examined in human cohorts to date.

Because AT is an anticoagulant, it has the potential to increase the risk of bleeding. Treatment with AT had a modest effect on the incidence of major bleeding events. The absolute increase in the incidence of major bleeding during the whole observation period was 1.8% in patients of the AT treatment group with DIC and 6.7% in patients without DIC. Thus, the absolute increase in the risk of major bleeding appeared to remain relatively low for AT-treated patients at least in the presence of DIC.

It is to be recognized that a significant limitation of the present analysis is that it was performed retrospectively in subgroups of patients that were not predefined by the primary analytic plan and therefore allows us only to generate hypotheses. However, as the results are biologically plausible and consistent with a considerable amount of background of previous experimental and clinical experience, they provide the rationale for new randomized controlled trials. These trials should address the hypothesis that treating DIC in severe sepsis with AT without concomitant administration of heparin may significantly reduce mortality.

Author contribution

Role of each author in the study: J. Kienast[1,3], M. Juers[1,3], C. J. Wiedermann[1,3], J. N. Hoffmann[3,4], H. Ostermann[3,4], R. Strauss[3,4], H-O. Keinecke[2,4], B.L. Warren[3–5], S. M. Opal[3,4,6]

[1]Prepared the manuscript draft; [2]performed the statistical analyses; [3]contributed to data assessment and interpretation; [4]performed critical review of multiple drafts of the manuscript; [5]co-organized the KyberSept trial and was responsible for the enrollment of patients from the South African centers; [6]principal investigator of the multinational KyberSept trial.

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Conflict of interest disclosure

This work was partly supported by a grant from ZLB Behring GmbH, Hattersheim, Germany. J. Kienast served as a medical advisor for ZLB Behring and has been an invited speaker at conferences supported by ZLB Behring. M. Juers is an employee of ZLB Behring and has stock options from the former Aventis Behring GmbH. C. J. Wiedermann has received honoraria for speaking for ZLB Behring; is a principal investigator of a ZLB Behring research grant at the Medical University of Innsbruck. H. Ostermann has received honoraria from ZLB Behring. R. Strauss has received honoraria and travel fees from ZLB Behring in 2003 and 2004. H-O. Keinecke is a statistical consultant for ZLB Behring. S. M. Opal received support from a clinical coordinating grant from Chiron. J. N. Hoffmann and B. L. Warren declare no conflicts of interest.

Appendix 1 DIC Definition of non-overt DIC

Criterion	Points
(1) Underlying disorder known to be associated with DIC	Yes = 2 points (assumed to be true for all patients in the KyberSept study)
(2) Baseline platelet count	
$\geq 100\,000\ \mu L^{-1}$	0 point
$< 100\,000\ \mu L^{-1}$	1 point
(3) Platelet count baseline	
> 24 h† platelet count	1 point
Platelet count baseline = 24 h† platelet count	0 point
Platelet count baseline < 24 h† platelet count	−1 point
(4) and (5) Baseline prothrombin time (PT)	
(4a) PT ≤ ULN* + 3 s	0 point
PT > ULN* + 3 s	1 point
(5a) PT baseline > 24 h† PT	−1 point
PT baseline = 24 h† PT	0 point
PT baseline < 24 h† PT	1 point
(4b) Quick test	
≥70%	0 point
< 70%	1 point
(5b) Quick at baseline < 24 h† Quick	−1 point
Quick at baseline = 24 h† Quick	0 point
Quick at baseline > 24 h† Quick	1 point
(4c) PT ratio/INR	
≤1.4	0 point
> 1.4	1 point
(5c) PT ratio/INR at baseline > 24 h† PT ratio/INR	−1 point
PT ratio/INR at baseline = 24 h† PT ratio/INR	0 point
PT ratio/INR at baseline < 24 h† PT ratio/INR	1 point
(6) Baseline D-dimer	
$\leq 0.39\ \mu g\ mL^{-1}$	0 point
> 0.39 to $\leq 4\ \mu g\ mL^{-1}$	2 points
$> 4\ \mu g\ mL^{-1}$	3 points
(7) Baseline AT III	
≥ 70%	0 point
< 70%	1 point
(8) Baseline protein C	
≥ 70%	0 point
< 70%	1 point
Sum: 1 + 2 + 3 + 4 (either a, b, or c) + 5 (either a, b, or c) + 6 + 7 + 8	< 7 points: No non-overt DIC ≥7 points: non-overt DIC

*Upper limit of center-specific normal range.
†Preferably at 24 h, but up to day 4 if not available before.

Appendix 2 Definition of overt DIC

Criterion	Points
(1) Baseline platelet count:	
$\geq 100\,000\ \mu L^{-1}$	0 point
$50\,000$ to $< 100\,000\ \mu L^{-1}$	1 point
$< 50\,000\ \mu L^{-1}$	2 points
(2) Baseline D-dimer	
$\leq 0.39\ \mu g\ mL^{-1}$	0 point
> 0.39 to $\leq 4\ \mu g\ mL^{-1}$	2 points
$> 4\ \mu g\ mL^{-1}$	3 points
(3a) PT	
≤ ULN* + 3 s	0 point
> ULN* + 3 s to ≤ULN* + 6 s	1 point
> ULN* + 6 s	2 points
(3b) Quick test	
≥ 70%	0 point
< 70% to ≥40%	1 point
< 40%	2 points
(3c) PT ratio/INR	
≤ 1.4	0 point
> 1.4 to ≤2.3	1 point
> 2.3	2 points
(4) Fibrinogen level	
$\geq 1\ g\ L^{-1}$ ($2.941\ \mu mol\ L^{-1}$)	0 point
$< 1\ g\ L^{-1}$ ($2.941\ \mu mol\ L^{-1}$)	1 point
Sum: 1 + 2 + 3 (either a, b, or c) + 4	≥5 points: overt DIC

*Upper limit of center specific normal range.

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